

Exp No:9) Matplotlib library in data science.

Aim:

To understand the use of the matplotlib library in data science through visualization.

Description:

Use Iris data set to understand the Matplot lib library

Code:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

data = pd.read_excel('/content/Social_Network_Ads.xlsx')

data
data.info()
data.describe()
data.value_counts('Gender')
dummies = pd.get_dummies(data.Gender)
FinalDataset = pd.concat(
    [dummies, data.iloc[:, [0, 2, 3, 4]]],
    axis=1
)
FinalDataset.head()
sns.scatterplot(
    x='Age',
    y='EstimatedSalary',
    hue='Gender',
    data=data
)
```

```
plt.show()

sns.scatterplot(
    x='Age',
    y='EstimatedSalary',
    hue='Purchased',
    data=data
)
```

```
plt.show()

g = sns.pairplot(
    data[['Age', 'EstimatedSalary', 'Purchased']],
    hue='Purchased',
    height=3
)
```

```
for ax in g.axes.flat:
    if ax is not None:
        ax.tick_params(axis='x', labelrotation=45)
```

```
plt.show()

sns.FacetGrid(
    data,
    hue='Gender',
    height=5
).map(
    sns.histplot,
    'EstimatedSalary'
).add_legend()
```

```
plt.show()
```

```

sns.FacetGrid(
    data,
    hue='Gender',
    height=5
).map(
    sns.histplot,
    'Age'
).add_legend()

plt.show()

import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

data = pd.read_excel('/content/Social_Network_Ads.xlsx')
data
data.info()
data.describe()
data.value_counts('Gender')
sns.countplot(x='Gender', data=data)
plt.show()

dummies = pd.get_dummies(data.Gender)

FinalDataset = pd.concat(
    [dummies, data.iloc[:, [0, 2, 3, 4]]],
    axis=1
)

FinalDataset.head()

```

```
sns.scatterplot(
    x='Age',
    y='EstimatedSalary',
    hue='Gender',
    data=data
)
plt.show()

sns.scatterplot(
    x='Age',
    y='EstimatedSalary',
    hue='Purchased',
    data=data
)
plt.show()

g = sns.pairplot(
    data[['Age', 'EstimatedSalary', 'Purchased']],
    hue='Purchased',
    height=3
)

for ax in g.axes.flat:
    if ax is not None:
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plt.show()

sns.FacetGrid(
    data,
    hue='Gender',
    height=5
```

```
).map(  
    sns.histplot,  
    'EstimatedSalary'  
).add_legend()
```

```
plt.show()  
  
sns.FacetGrid(  
    data,  
    hue='Gender',  
    height=5
```

```
).map(  
    sns.histplot,  
    'Age'  
).add_legend()
```

```
plt.show()  
  
import pandas as pd  
import numpy as np  
import seaborn as sns  
import matplotlib.pyplot as plt  
  
data = pd.read_excel('/content/Social_Network_Ads.xlsx')  
  
data  
  
data.info()  
  
data.describe()  
  
data.value_counts('Gender')  
  
sns.countplot(x='Gender', data=data)  
  
plt.show()  
  
dummies = pd.get_dummies(data.Gender)
```

```
FinalDataset = pd.concat(  
    [dummies, data.iloc[:, [0, 2, 3, 4]]],  
    axis=1  
)
```

```
FinalDataset.head()
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```
sns.scatterplot(  
    x='Age',  
    y='EstimatedSalary',  
    hue='Gender',  
    data=data  
)
```

```
plt.show()
```

```
sns.scatterplot(  
    x='Age',  
    y='EstimatedSalary',  
    hue='Purchased',  
    data=data  
)
```

```
plt.show()
```

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g = sns.pairplot(  
    data[['Age', 'EstimatedSalary', 'Purchased']],  
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sns.FacetGrid(
    data,
    hue='Gender',
    height=5
).map(
    sns.histplot,
    'EstimatedSalary'
).add_legend()
```

```
plt.show()

sns.FacetGrid(
    data,
    hue='Gender',
    height=5
).map(
    sns.histplot,
    'Age'
).add_legend()
```

```
plt.show()
```

Output:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 400 entries, 0 to 399

Data columns (total 5 columns):

```
# Column      Non-Null Count  Dtype
```

```
---  -----
```

```
0  User ID      400 non-null  int64
```

```
1  Gender       400 non-null  object
```

```
2  Age          400 non-null  int64
```

```
3  EstimatedSalary 400 non-null  int64
```

```
4  Purchased     400 non-null  int64
```

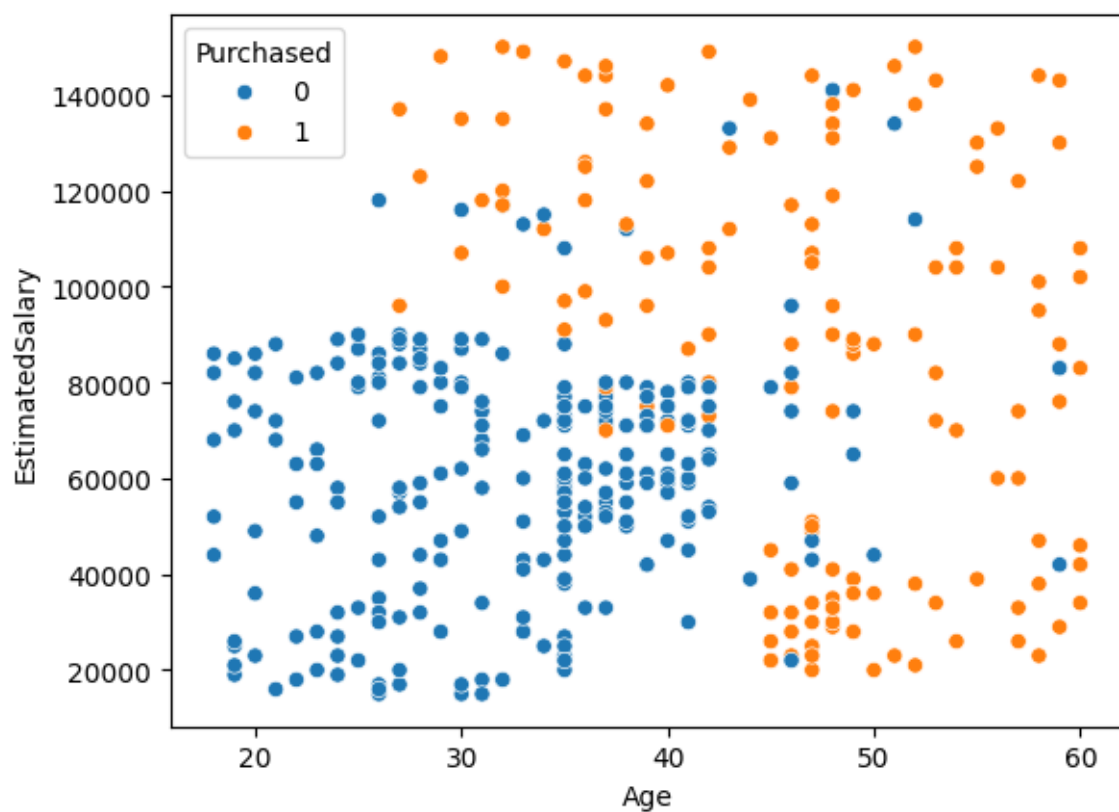
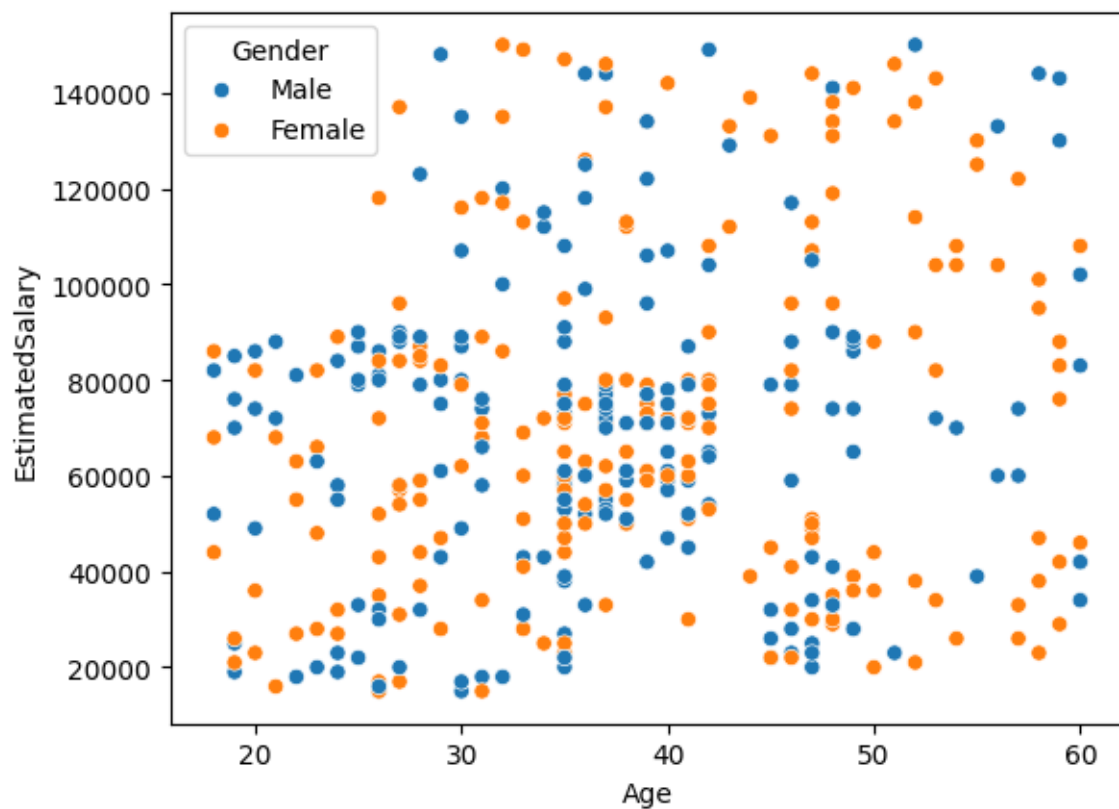
```
dtypes: int64(4), object(1)
```

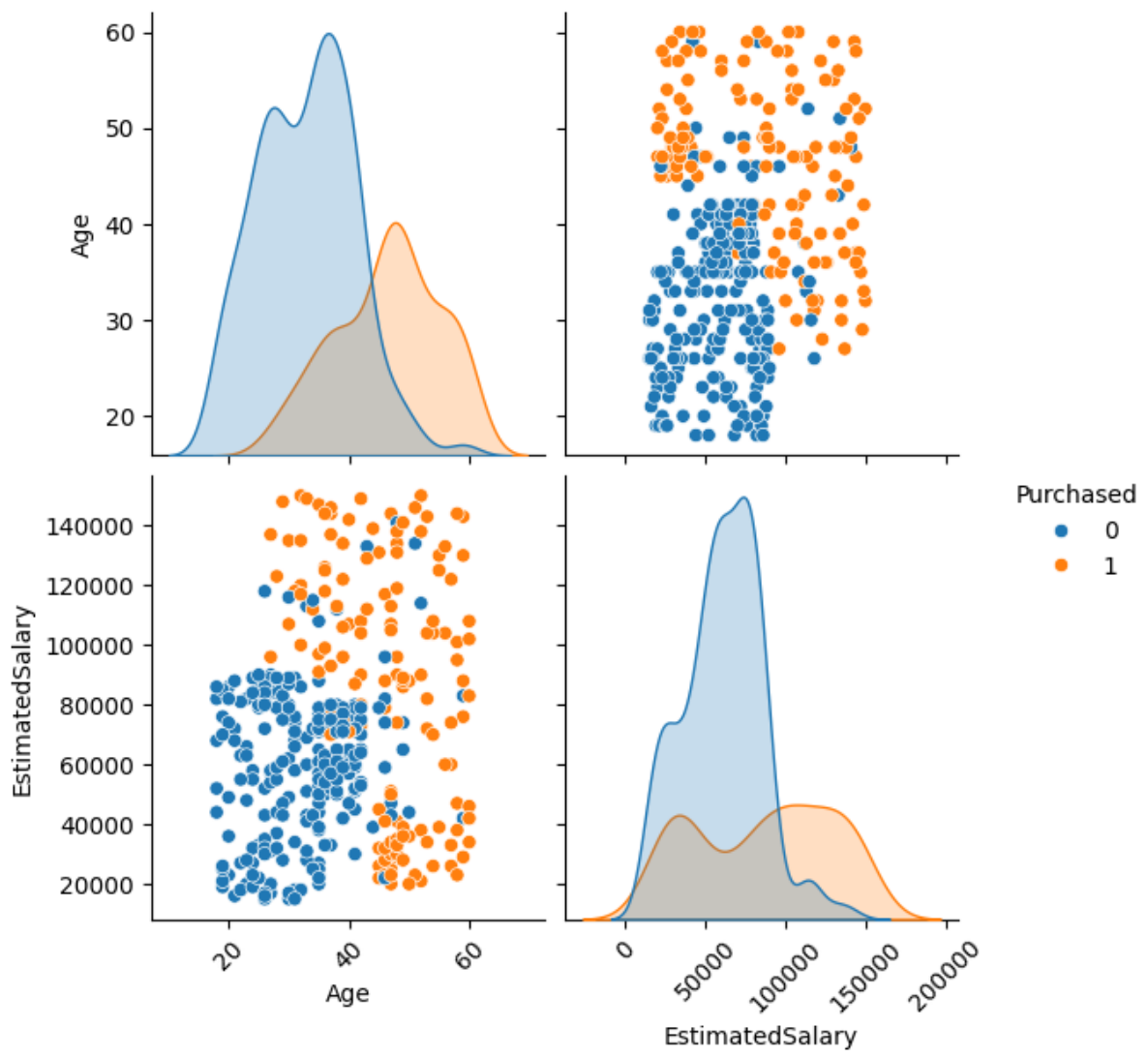
memory usage: 15.8+ KB

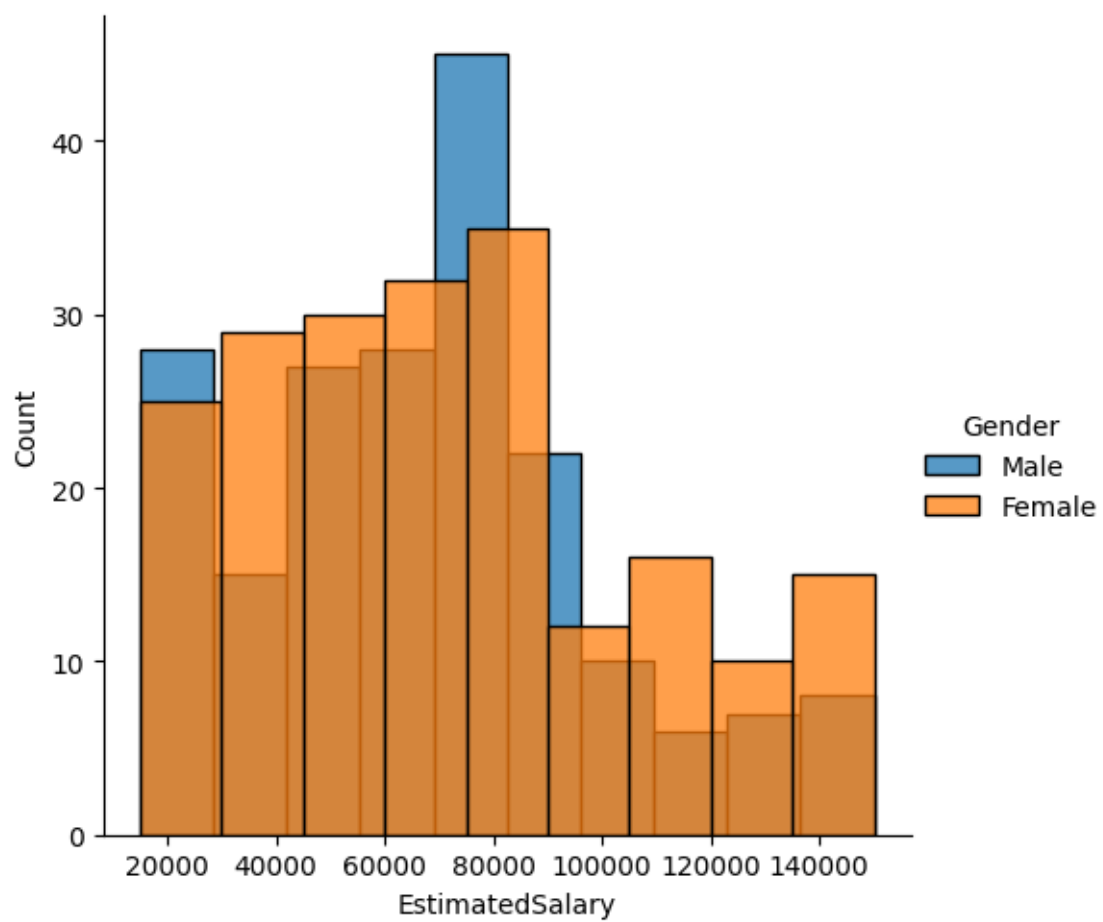
	User ID	Age	EstimatedSalary	Purchased
count	4.000000e+02	400.000000	400.000000	400.000000
mean	1.569154e+07	37.655000	69742.500000	0.357500
std	7.165832e+04	10.482877	34096.960282	0.479864
min	1.556669e+07	18.000000	15000.000000	0.000000
25%	1.562676e+07	29.750000	43000.000000	0.000000
50%	1.569434e+07	37.000000	70000.000000	0.000000
75%	1.575036e+07	46.000000	88000.000000	1.000000
max	1.581524e+07	60.000000	150000.000000	1.000000
	count			
Gender				
Female	204			
Male	196			

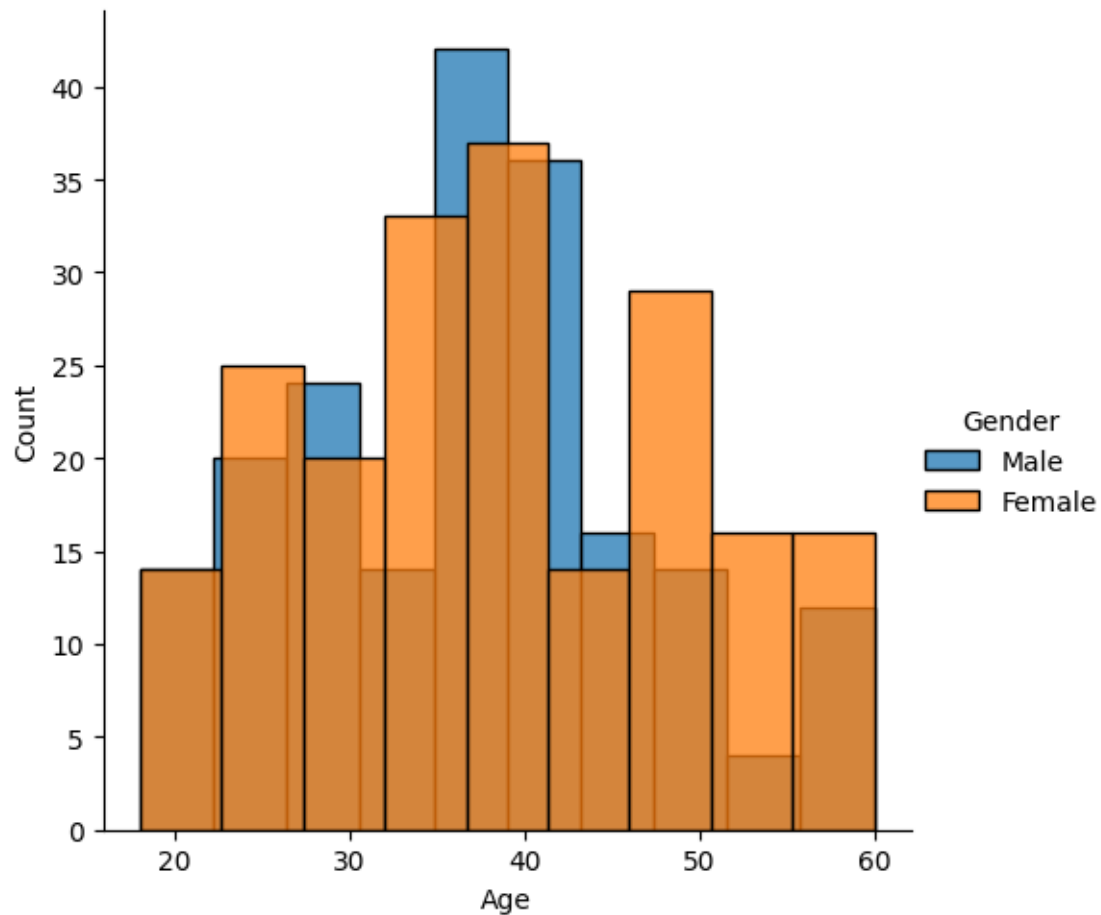
dtype: int64

	Female	Male	User ID	Age	EstimatedSalary	Purchased
0	False	True	15624510	19	19000	0
1	False	True	15810944	35	20000	0
2	True	False	15668575	26	43000	0
3	True	False	15603246	27	57000	0
4	False	True	15804002	19	76000	0









```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 400 entries, 0 to 399
```

```
Data columns (total 5 columns):
```

```
#   Column      Non-Null Count  Dtype
```

```
---  ---
```

```
0  User ID    400 non-null  int64
```

```
1  Gender     400 non-null  object
```

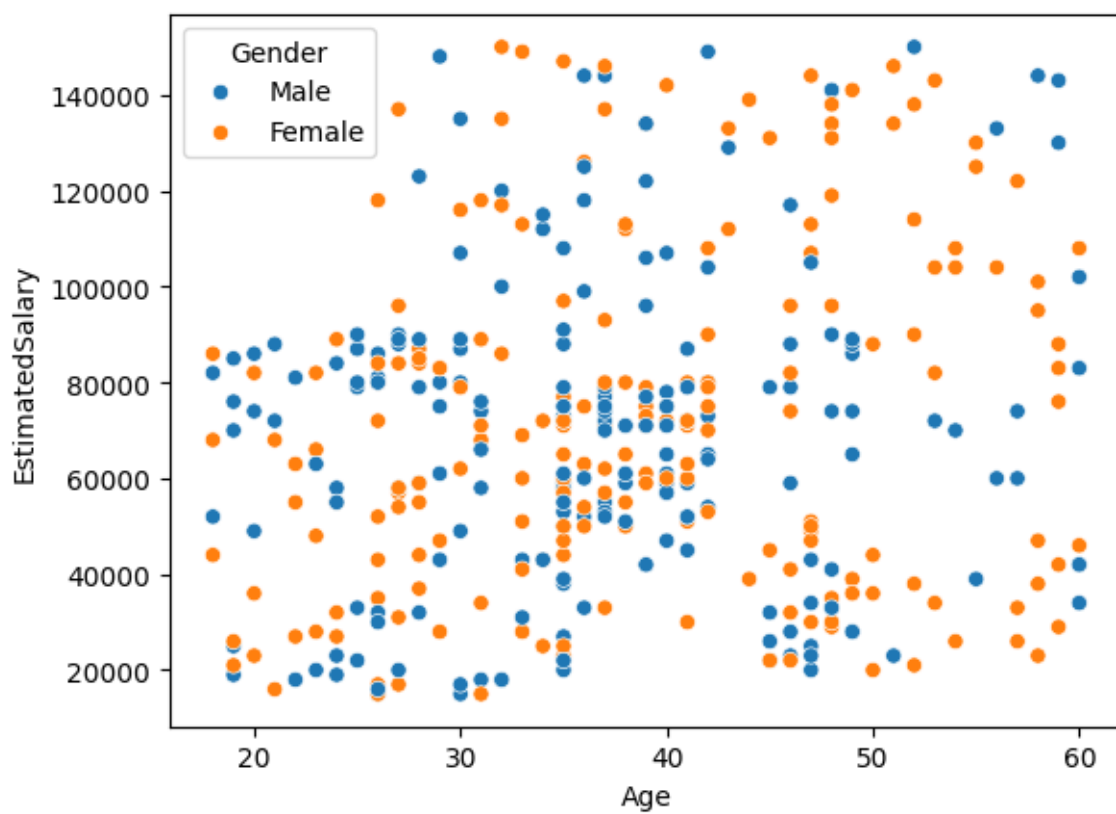
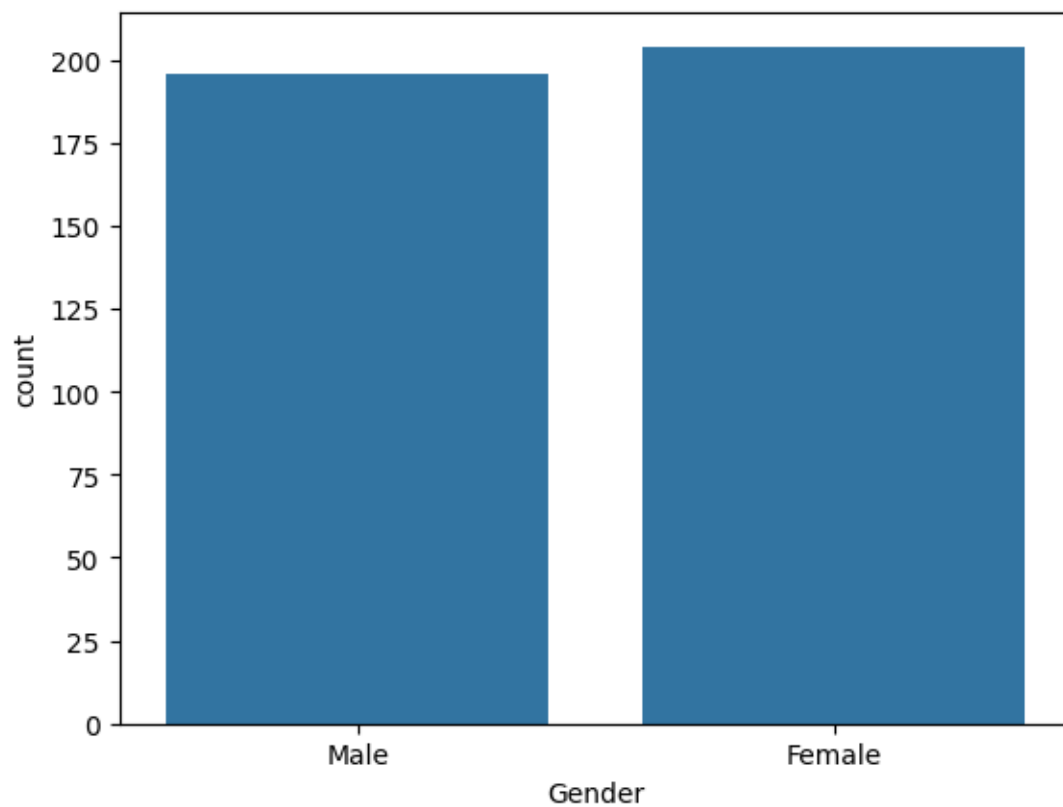
```
2  Age        400 non-null  int64
```

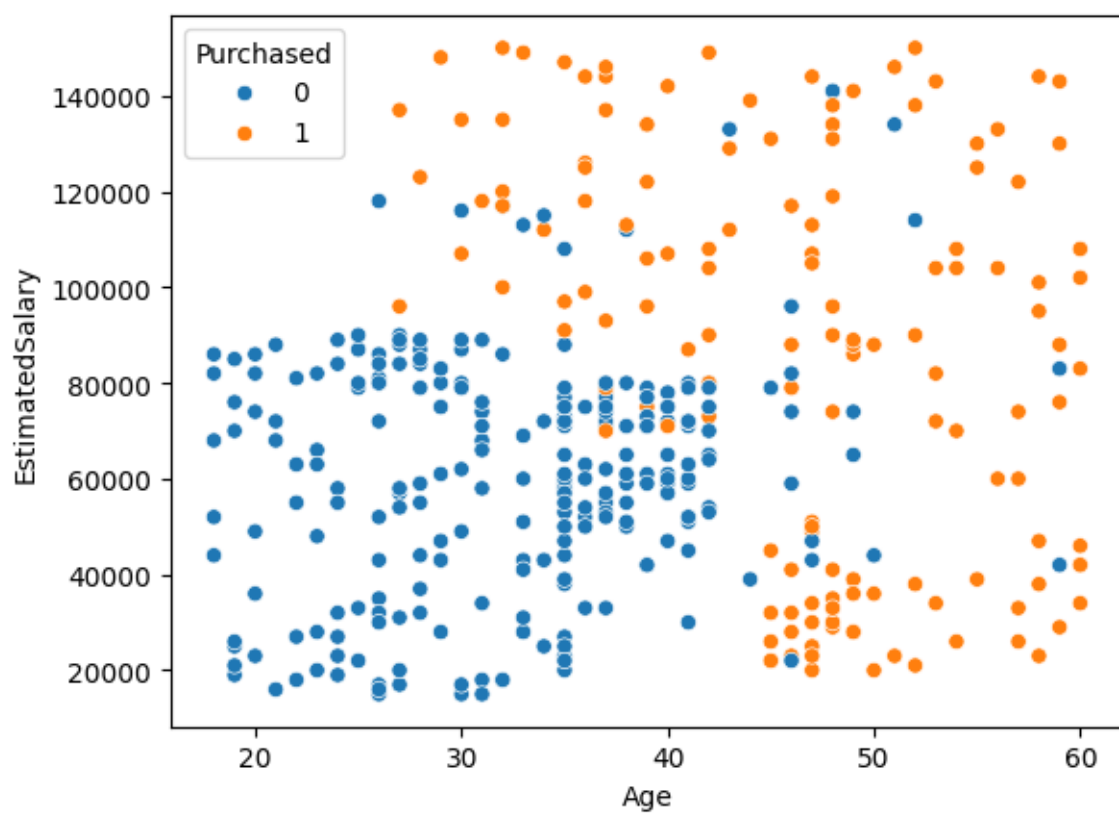
```
3  EstimatedSalary 400 non-null  int64
```

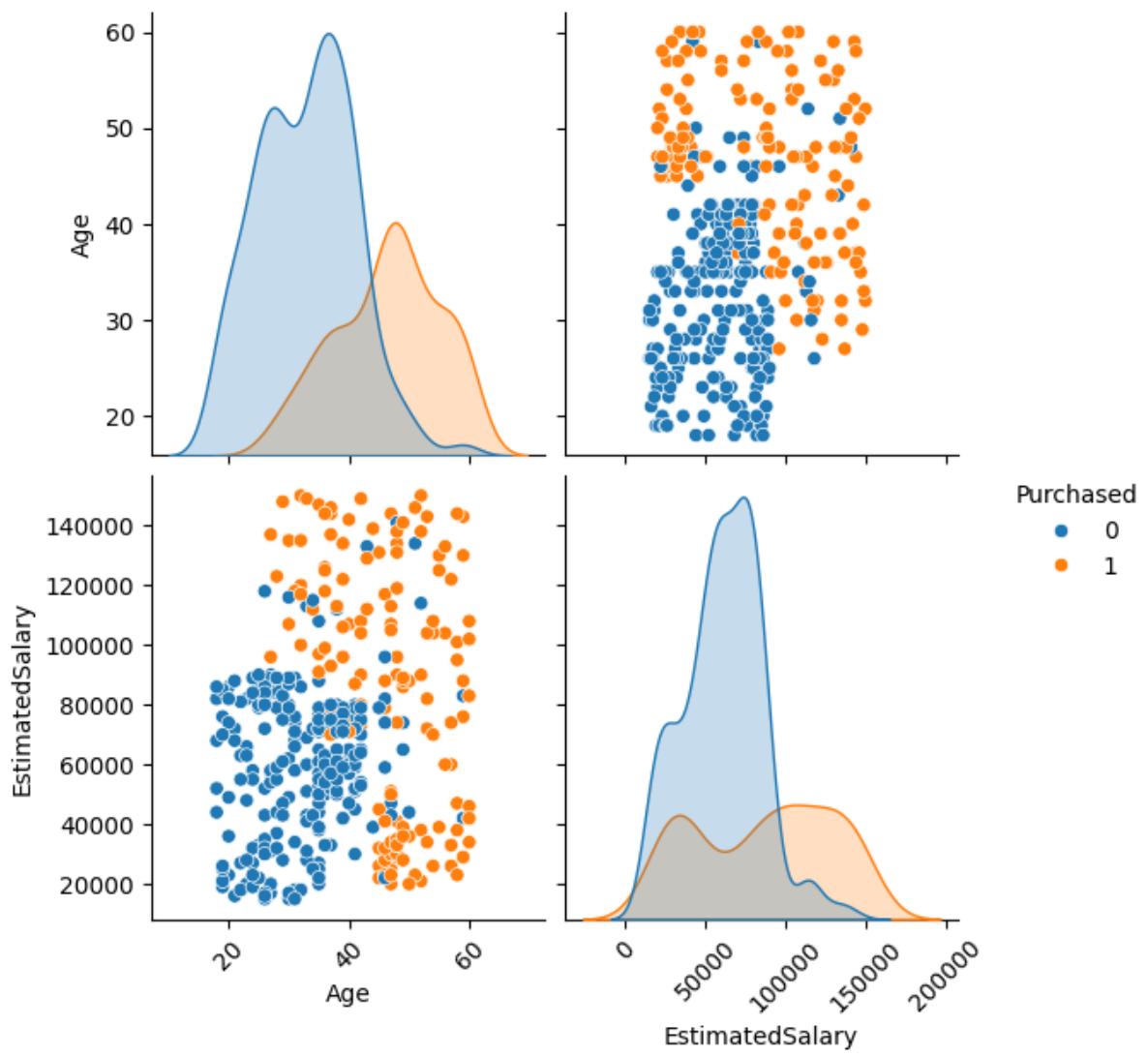
```
4  Purchased   400 non-null  int64
```

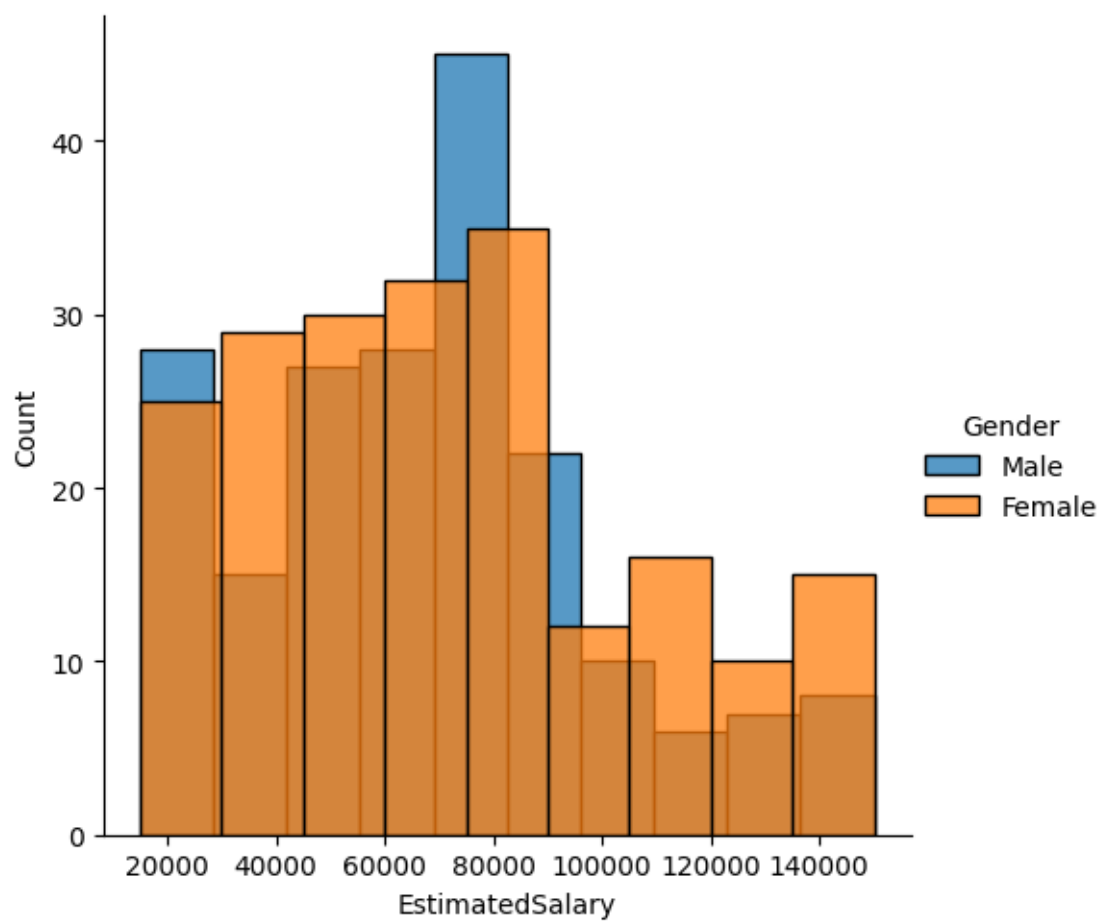
```
dtypes: int64(4), object(1)
```

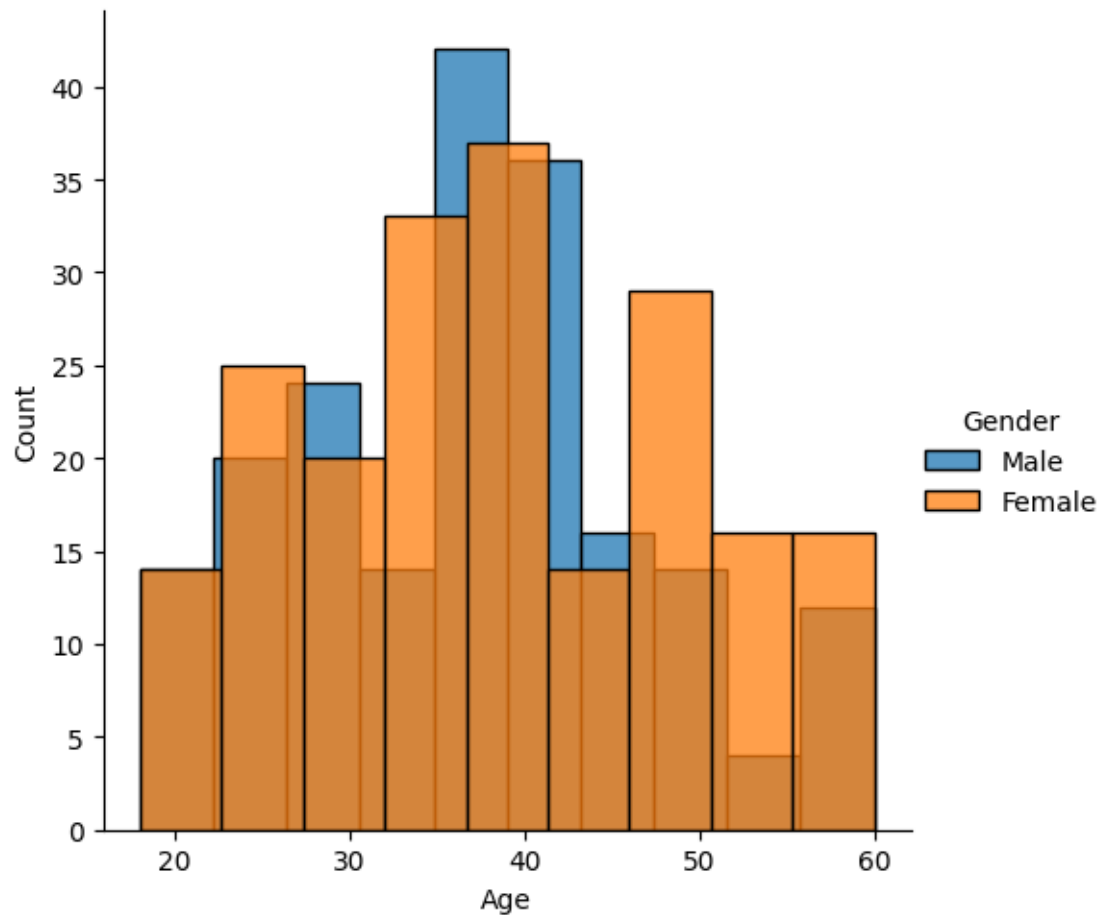
```
memory usage: 15.8+ KB
```











```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 400 entries, 0 to 399
```

```
Data columns (total 5 columns):
```

```
#   Column      Non-Null Count  Dtype
```

```
---  ---
```

```
0  User ID    400 non-null  int64
```

```
1  Gender     400 non-null  object
```

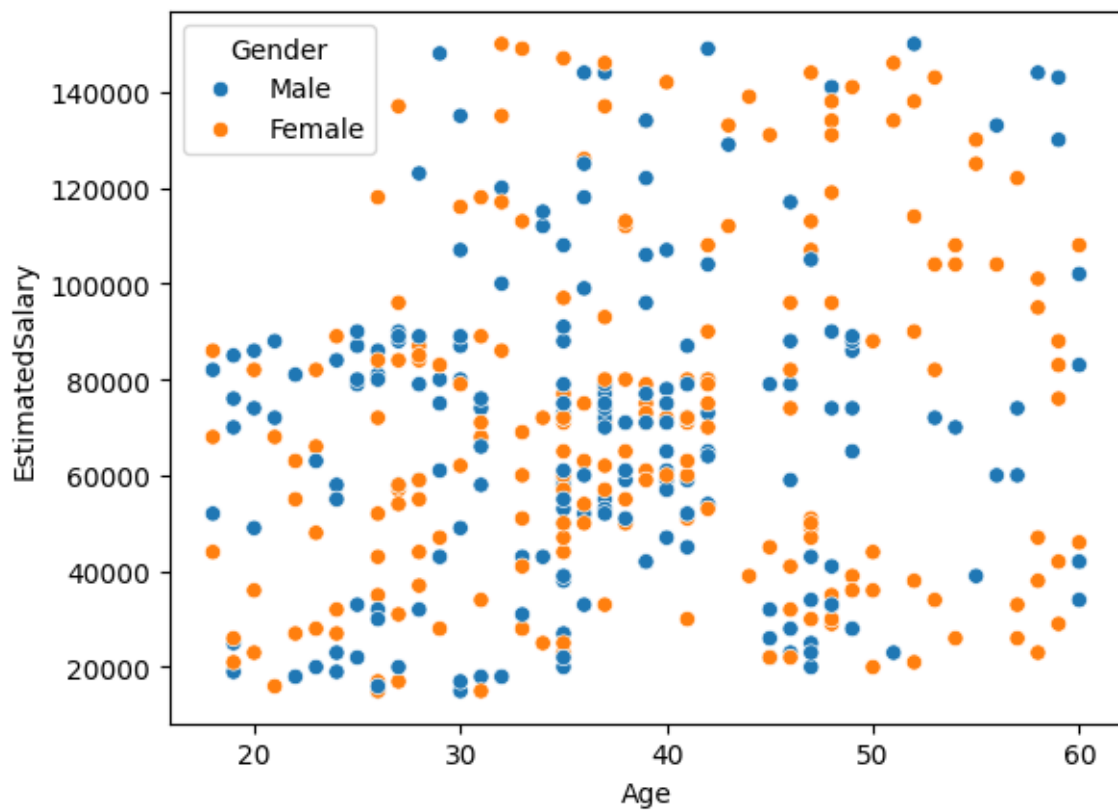
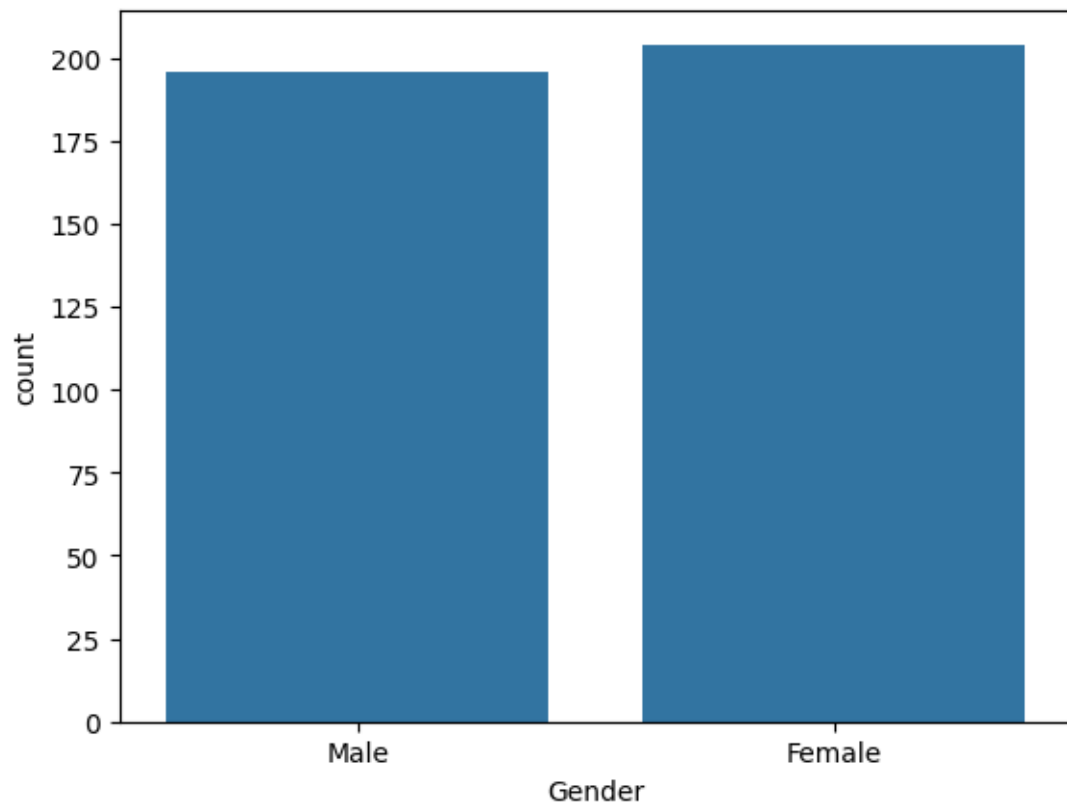
```
2  Age        400 non-null  int64
```

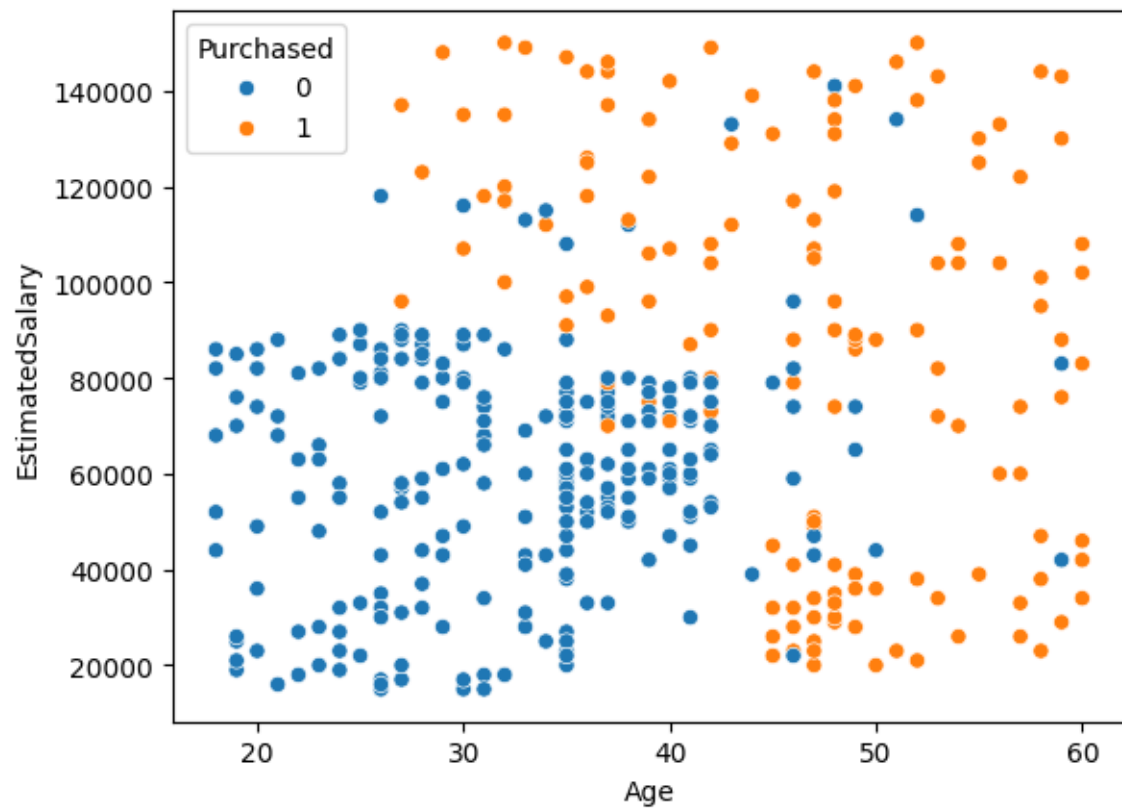
```
3  EstimatedSalary 400 non-null  int64
```

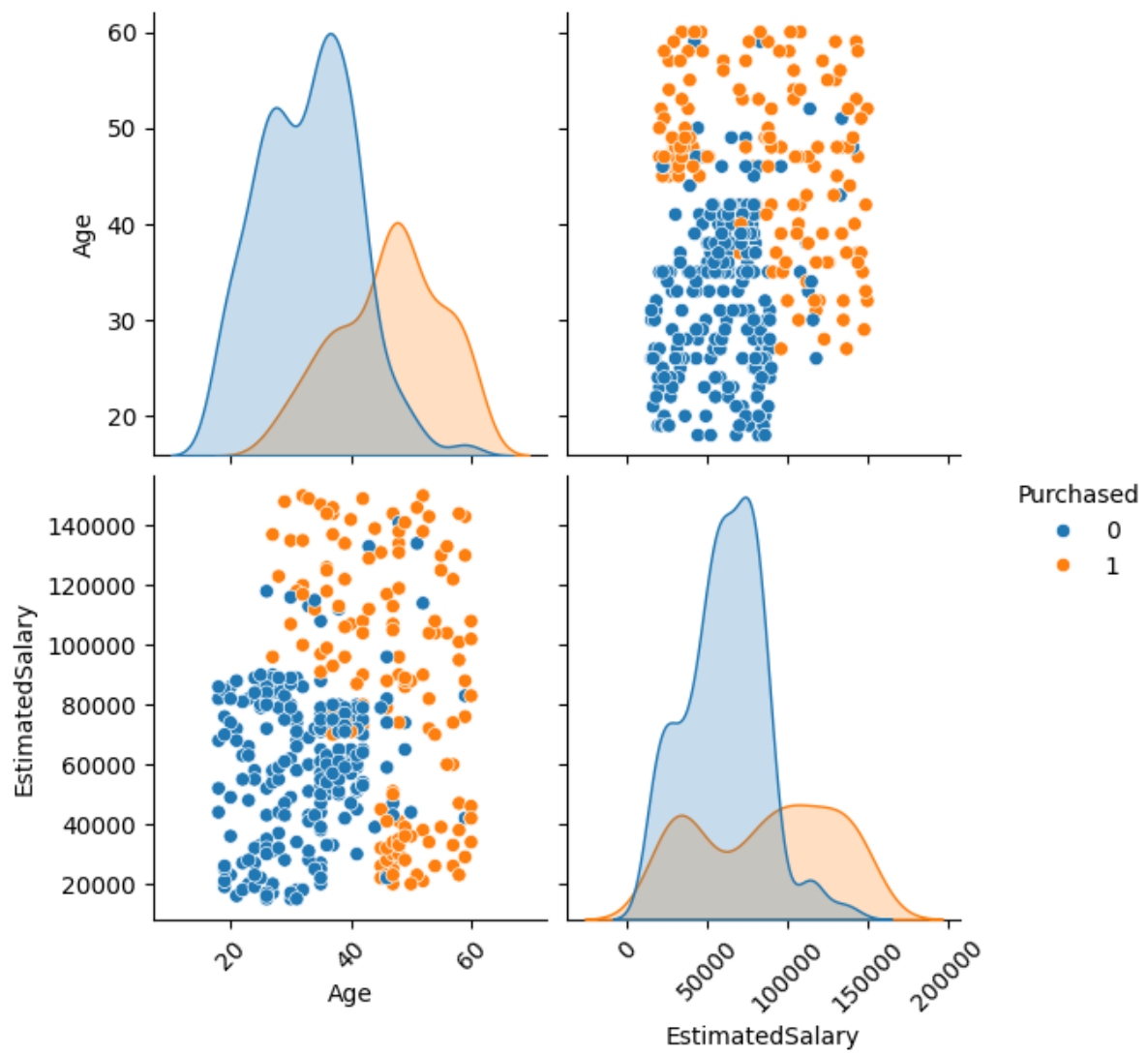
```
4  Purchased   400 non-null  int64
```

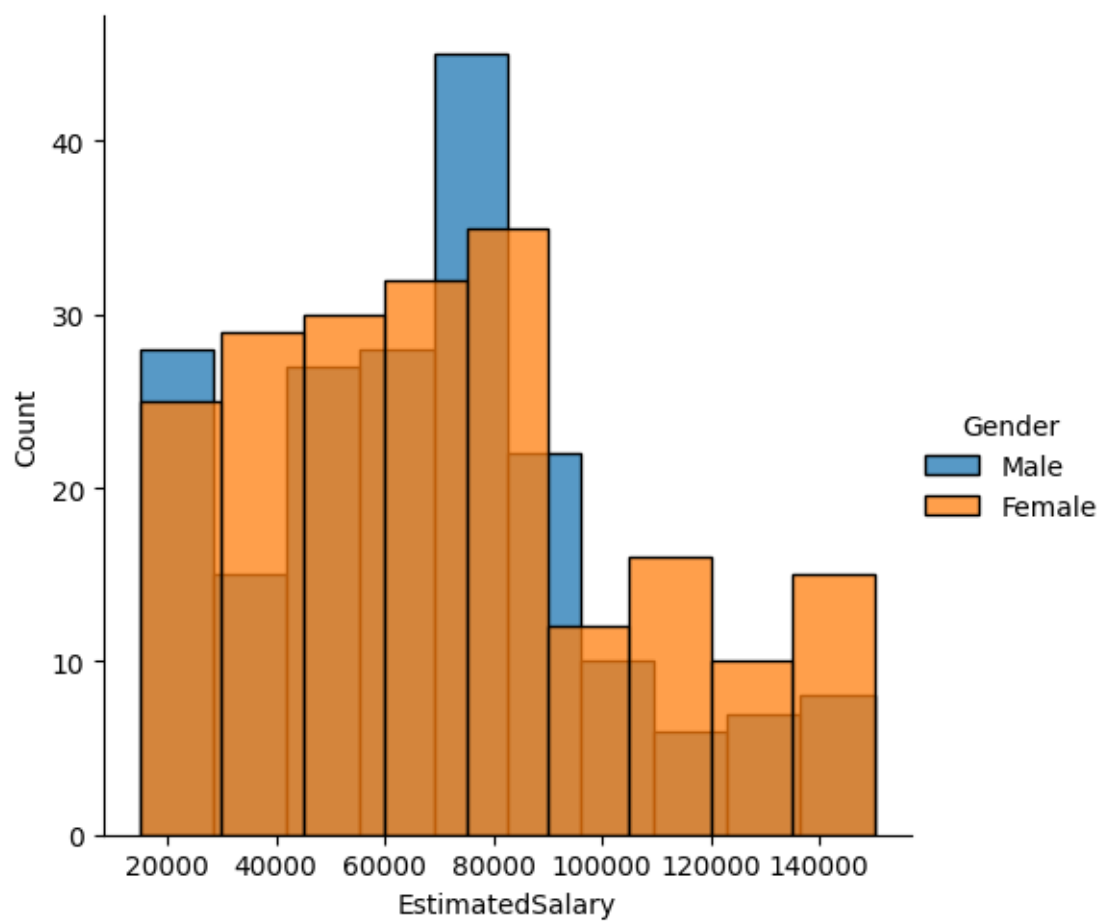
```
dtypes: int64(4), object(1)
```

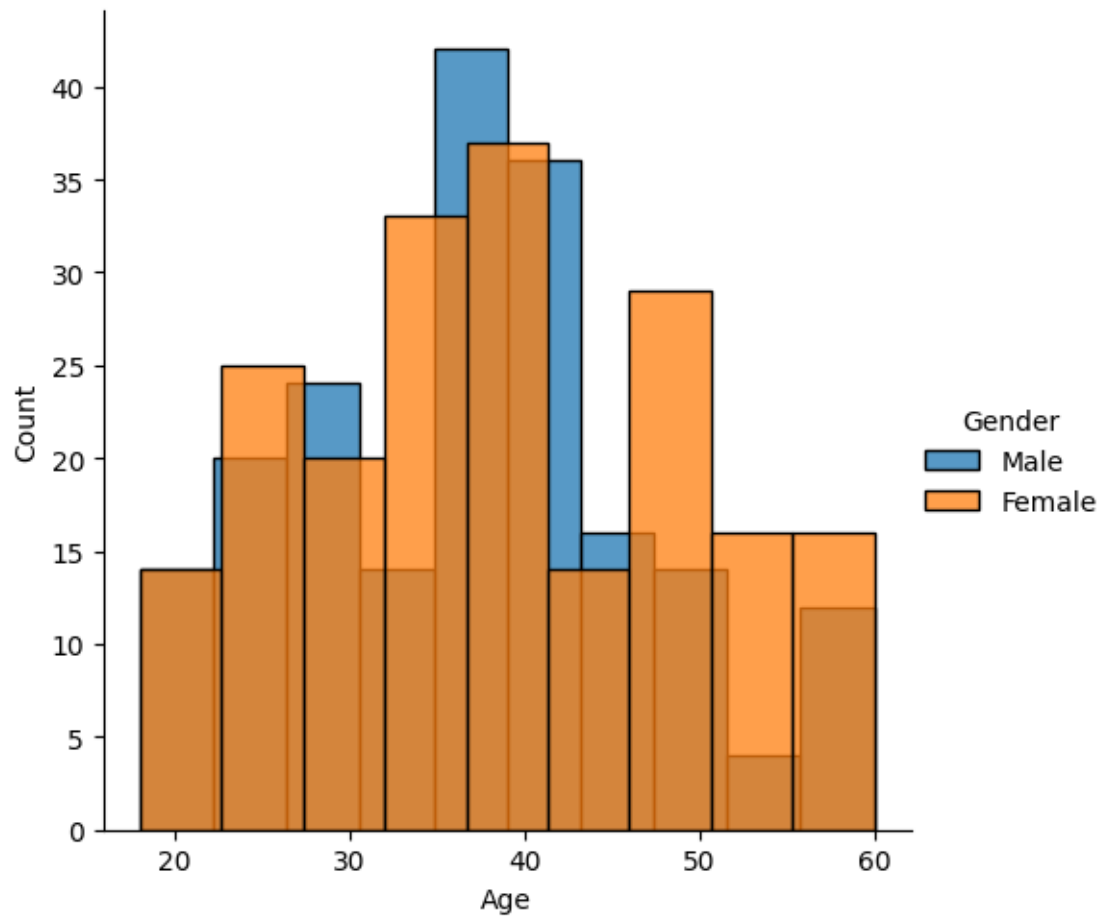
```
memory usage: 15.8+ KB
```











Result:

The iris dataset was successfully visualized using different plots, helping to understand the distributions and relationships among the data features.